

Lab Exercise 16

HIGH LEVEL PROGRAMMING EXTENSIONS – PROCEDURES

AIM

To implement **stored procedures** in MySQL using the `Vegetable` table.

PROCEDURE

1. Select the `VegetableDB` database.
2. Create a procedure to display all records from the `Vegetable` table.
3. Execute the procedure using the `CALL` command.
4. Create a procedure that accepts a vegetable ID and returns its price category.
5. Execute the procedure by passing the required parameters.

Problem 1: Display All Vegetable Records

Create Procedure

```
USE VegetableDB;

DELIMITER //

CREATE PROCEDURE GetVegetables()
BEGIN
    SELECT * FROM Vegetable;
END //

DELIMITER ;
```

Execute Procedure

```
CALL GetVegetables();
```

```
mysql> CALL GetVegetables();
```

VegID	VegName	Category	Price	Quantity
1	Tomato	Vegetable	30.50	100
2	Potato	Vegetable	25.00	200
3	Carrot	Root	40.00	80
4	Onion	Vegetable	35.00	150
5	Beetroot	Vegetable	45.50	90
6	Cabbage	Leafy	50.00	70
10	Spinach	Leafy	28.00	60
11	Beetroot	Root	42.00	50

8 rows in set (0.00 sec)

Problem 2: Get Vegetable Price Level

Instead of **customer level**, we use **vegetable price level**.

Logic

- Price $\leq 30 \rightarrow$ **LOW**
- Price $\leq 45 \rightarrow$ **MEDIUM**
- Price $> 45 \rightarrow$ **HIGH**

Create Procedure

```
DELIMITER //
```

```
CREATE PROCEDURE GetVegetableLevel(  
    IN p_vegid INT,  
    OUT p_level VARCHAR(20)  
)  
BEGIN  
    DECLARE v_price DECIMAL(10,2);  
  
    SELECT Price  
    INTO v_price  
    FROM Vegetable  
    WHERE VegID = p_vegid;  
  
    IF v_price <= 30 THEN  
        SET p_level = 'LOW';  
    ELSEIF v_price <= 45 THEN  
        SET p_level = 'MEDIUM';  
    ELSE  
        SET p_level = 'HIGH';  
    END IF;  
END //
```

```
DELIMITER ;
```

Execute Procedure

For **VegID = 1 (Tomato)**:

```
CALL GetVegetableLevel(1, @level);  
SELECT @level AS Vegetable_Level;
```

```
mysql> SELECT @level AS Vegetable_Level;  
+-----+  
| Vegetable_Level |  
+-----+  
| MEDIUM        |  
+-----+
```

For **VegID = 5 (Brinjal)**:

```
CALL GetVegetableLevel(5, @level);  
SELECT @level AS Vegetable_Level;
```

```
mysql> SELECT @level AS Vegetable_Level;  
+-----+  
| Vegetable_Level |  
+-----+  
| HIGH           |  
+-----+  
1 row in set (0.00 sec)
```